SOKOINE UNIVERSITY OF AGRICULTURE
DEPARTMENT OF AGRICULTURAL ENGINEERING

ABE 715: COMPUTER PROGRAMMING

Date Given: Friday, 13/02/2026 Submission Date: Friday, 20/02/2026
Student Name: _____ Reg. No: _____

INSTRUCTIONS:

1. This assignment consists of THREE sections: A, B, and C.
2. Answer ALL questions. This is a Take-Home Assignment.
3. Submit your solutions as a single PDF report containing your Python code snippets and the resulting outputs/plots.
4. Total Marks: 100.

SECTION A: COMPUTATIONAL CONCEPTS (10 Marks)

Briefly answer the following questions regarding Python libraries used in Irrigation Engineering. (2.5 Marks each)

1. Vectorization: Explain why agricultural engineers prefer using NumPy arrays over standard Python Lists when processing large datasets of soil moisture readings.
2. DataFrames: In the Pandas library, what is the primary difference between a Series and a DataFrame? How is this useful for handling time-series weather data?
3. Numerical Solving: Why are numerical methods (like Newton-Raphson) required to solve the Colebrook-White equation for pipe friction factors, whereas Darcy-Weisbach can be solved analytically?
4. IoT Protocols: In an automated irrigation system, why would you choose the MQTT protocol over standard HTTP for communication between soil sensors and the central server?

SECTION B: ALGORITHMIC SOLUTIONS (20 Marks)

Question 5: Normal Depth Calculation (10 Marks)
(Problem 1: Open Channel Flow)

You are designing a trapezoidal irrigation canal. The flow rate (Q) is governed by Manning's Equation, which is implicit for normal depth (y_n):

$$Q = \frac{1}{n}\frac{(A)^{5/3}}{(P)^{2/3}}\sqrt{S_0}$$

Where Area $A = (b + zy)y$ and Wetted Perimeter $P = b + 2y\sqrt{1 + z^2}$.

Task: Write a Python script using scipy.optimize.fsolve (or write your own Newton-Raphson function) to find the Normal Depth (y_n).

- Given: Flow $Q = 15\,m^3/s$, Slope $S_0 = 0.001$, Roughness $n = 0.015$.
- Geometry: Bottom width $b = 3.0\,m$, Side slope $z = 2$.
- Output: Print the calculated depth in meters to 3 decimal places.

Question 6: Reservoir Volume Estimation (10 Marks)
(Real World Problem 2: Hydrology Integration)

An automated flow meter at a weir station records the inflow rate (m^3/s) into a reservoir every hour during a 10-hour storm event.

Data: flow_rates = [5, 12, 25, 40, 35, 20, 15, 8, 4, 2, 0]
(Measurements taken at $t = 0, 1, 2...10$ hours).

Task:

(a) Write a Python function to calculate the Total Volume of water (m^3) entering the reservoir using Simpson's 1/3 Rule.

(b) Ensure you handle unit conversion correctly (Hours to Seconds).

(c) Print the total volume in cubic meters.

SECTION C: INTEGRATED ENGINEERING SYSTEMS (70 Marks)

Question 7: Climate Data Pipeline: ERA5 & PyETo (40 Marks)
(Problem 3: Geospatial Data & APIs)

You are tasked with analyzing the water requirements for a farm in Morogoro (Lat: -6.82, Lon: 37.66) for January 2025. Instead of static data, you must retrieve raw climate data and compute parameters.

Task: Write a Python script to:

(a) Data Retrieval: Use the cdsapi library to fetch ERA5 reanalysis data for the specified location and time period. Retrieve: Rainfall, Temperature (Min/Max), Solar Radiation, and Wind Speed.

(b) ETo Calculation: Use the pyeto library to calculate the daily FAO-56 Reference Evapotranspiration (ETo). (Note: Ensure unit conversions, e.g., Kelvin to Celsius, are handled correctly).

(c) Deficit Analysis: Create a Pandas DataFrame with columns: ['Date', 'Rainfall_mm', 'ETo_mm', 'Deficit']. Calculate $Deficit = ETo - Rainfall$.

(d) Visualization: Plot a dual-axis chart (using Matplotlib) showing Daily Rainfall (Bar) and calculated ETo (Line).

Question 8: Crop Yield Prediction (15 Marks)
(Problem 4: Machine Learning)

A research station has historical data relating Water Application (mm) and Nitrogen (kg/ha) to Maize Yield (tons/ha).

Training Data (X): [[300, 50], [400, 80], [500, 100], [600, 120], [350, 60]]
Yield Labels (y): [2.5, 4.0, 5.5, 6.2, 3.1]

Task:

(a) Use sklearn to train a Linear Regression model on this data.

(b) Predict the yield for a new season where Water = 450mm and Nitrogen = 90kg/ha.

(c) Print the model coefficients (Sensitivity of yield to water vs fertilizer).

Question 9: Smart Greenhouse Control Logic (15 Marks)
(Problem 5: IoT & Automation)

You are programming an ESP32 controller for a greenhouse. The system reads Temperature (T) and Soil Moisture (SM) every minute.

Task: Write the Python Logic (Pseudocode or Python implementation) for a control loop that:

(a) Checks sensor bounds (Defensive Programming): Ensure T is between 0-50°C and SM is 0-100%. If not, log an "Error".

(b) Implements Hysteresis Control for the Exhaust Fan:

- Turn Fan ON if $T > 30°C$.
- Turn Fan OFF only if $T < 26°C$ (to prevent rapid toggling).

(c) Triggers an SMS Alert (print "ALERT") if $SM < 20\%$ (Critical Wilting Point).